

Jawaharlal Nehru Govt. Engineering College, Sundernagar, Distt. Mandi(H.P)
Department of Computer Science and Engineering

Program:	B.Tech	Semester:	3rd
Subject Code:	CSPC-311	Subject Title:	Data Structure and Algorithm
MST No.:	2nd	Course Coordinator:	Rahul Pal Singh
Maximum Marks:	50	Time Duration:	2hrs.
Date of MST:	07-11-2024 (Evening)	Roll Number:	

Note: All Questions are Compulsory

Q. No.	Questions	CO/Marks
Q1.	Do the following (Give algorithms): 1. Find the validity of the expression using stack data structure: $(4+6*(4+3)*(7*3)+5)$ 2. Using Stack data structure find out the Postfix notation of the expression: $4+6*4+3*7*3+5$	CO2, L3 5+5=10
Q2.	Define AVL Tree? What are the properties of an AVL tree? Construct an AVL Tree from an empty Tree by inserting the following nodes (specify the balancing factor, height of a node, rotations required, updating the height of nodes after every insertion): 34, 72, 92, 75, 82, 100, 110, 14, 32, 16, 12, 7	CO3, L3 15
Q3.	Write an algorithm to implement Priority Queue using arrays. Find its time complexity? List applications of a queue.	CO2, L2 10
Q4.	Explain binary search tree? Write algorithms for Inserting and Deleting a node in BST. Analyse their time complexity from the worst case perspective. Do the following: a. Insert nodes 23, 56, 67, 35, 79, 64, 12, 7, 99, 80 on a binary search tree. b. Delete the nodes 79, 12 and 23 from the tree obtained in part (a)	CO3, L3 15

Course Outcomes(CO):

Compare the performance and analyse the complexity of various data structures

Able to design and implement Linear data structures like linked lists, queues, stacks, etc.

Able to design and implement Non-Linear data structures i.e. binary search tree, heap trees, etc

Able to perform operations on graph data structure

Learn & Compare algorithmic techniques to implement various sorting algorithms

Understand and apply the concept of hashing and hash table operations.

Level Number	L1	L2	L3	L4	L5	L6
Level Name	Remember	Understand	Apply	Analyze	Evaluate	Create